

COMPONENT-SPECIFIC TROUBLESHOOTING THEORY

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

ANS:

RAM Reseating Procedure

1. Shut down the computer and **disconnect the power cable**.
2. Open the **side panel/back cover** of the computer.
3. Locate the **RAM slots on the motherboard**.
4. Push the locking clips on both sides of the RAM slot outward.
5. Carefully remove the RAM module by holding its **edges**.
6. Check the RAM and slot for **dust or visible damage**.
7. Align the notch on the RAM with the slot and press it firmly into place.
8. Make sure both locking clips **click into position**.
9. Close the computer cover and reconnect the power cable.
10. Start the computer and check that it **boots normally**.

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.

Hint: Think about display issues, driver checks, and physical connections.

ANS:

GPU Troubleshooting Checklist

- ☐ **1. Identify the symptom** – Check for no display, screen flickering, artifacts, crashes, or poor gaming performance.
- ☐ **2. Check monitor and cable** – Make sure the monitor is powered on and the HDMI/DisplayPort cable is properly connected to the **GPU**, not the motherboard.
- ☐ **3. Check GPU physical connection** – Turn off and unplug the PC, then ensure the GPU is firmly seated in the PCIe slot.
- ☐ **4. Check GPU power cables** – Make sure the required **6-pin/8-pin PCIe power connectors** are securely connected.

- ☐ **5. Check the GPU driver** – Verify that the graphics driver is installed correctly and update/reinstall it if necessary.
- ☐ **6. Check cooling** – Make sure the GPU fans are working and the card is not overheating due to blocked airflow or dust.
- ☐ **7. Confirm the fix** – Restart the PC and run a game or graphics test to confirm that the display is stable and no artifacts or crashes occur.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

ANS:

Storage Drive Reconnection Test

Steps:

1. Shut down the test PC and disconnect the power.
2. Disconnect the **SATA data/power cable** from the SSD/HDD. *(For NVMe, remove the drive from its M.2 slot.)*
3. Start the PC and enter the **BIOS/UEFI**.
4. Check the storage/boot section — the drive should be **missing/not detected**. A message such as **“No Boot Device Found”** may also appear.
5. Shut down the PC and reconnect the SATA cables or reseal the NVMe drive.
6. Start the PC and check the BIOS again.
7. The drive should now appear in the **Storage/Boot Device list**, and the operating system should boot normally.

Observation:

Before reconnecting: Storage drive was not detected and the system showed a boot-device error.

After reconnecting: Drive was detected in BIOS and the system booted normally.

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

ANS:

Random PC Restart – Troubleshooting

Component to check first: RAM

Reason: Faulty or improperly seated RAM can cause system instability, crashes, and unexpected restarts, so it is a useful first hardware check.

Diagnostic Steps

1. **Shut down and unplug** the PC.
2. Open the cabinet and **locate the RAM modules**.
3. Remove and **resseat the RAM** firmly in the slot.
4. If there are multiple RAM sticks, **test one stick at a time** to identify a faulty module.
5. Check the RAM and slots for **dust or visible damage**.
6. Start the PC and check whether the random restarts continue.
7. If the problem remains, proceed to check the **GPU and storage drive** next.